Preface

The objective of this paper is to present a thorough analysis and forecast model of the trends in the air quality index (AQI) for the five years between 2018 and 2022. The main goal is to comprehend the trends in the AQI data and create a trustworthy model for AQI forecasting in 2023.

To do this, daily AQI data was gathered and painstakingly cleaned, taking care of any null values to guarantee the dataset's integrity and completeness. To find underlying trends, patterns, and anomalies in the data, a thorough exploratory data analysis (EDA) was carried out after the data cleaning procedure.

Then, five prediction models—'Exponential Smoothing', 'Extreme Gradient Boosting' (XGBoost), 'Bidirectional Long Short-Term Memory' (BiLSTM), 'Long Short-Term Memory' (LSTM), and 'Recurrent Neural Network' (RNN)—were created and assessed. To determine which model was the most accurate predictor, each model underwent extensive testing, and the results were compared.

The best-performing model was chosen and used to anticipate the AQI for 2023 based on this examination. To enhance air quality and protect public health, the results of this prediction study are meant to help decision-makers in government, environmental organizations, and public health.

The report is organized as follows: the procedure for gathering and cleansing data is described in the first part. We explore the exploratory data analysis in the second section. The creation and assessment of the prediction models are covered in detail in the third part. The AQI projections for 2023 are shown in the fourth part, and conclusions and suggestions are provided in the last section in light of the data.